TANVI UDAY KANCHAN

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SUMMARY

Cloud and software engineer with experience in AWS infrastructure, Kubernetes-based platforms, backend API development, and automation. Skilled in building cloud-native systems using Docker, EKS, Terraform, CloudFormation, and CI/CD pipelines, with additional experience in Databricks, PySpark, and Delta Lake for scalable data processing and analytics. Strong foundation in Linux, distributed systems, observability, and production support, with a focus on secure, reliable, and scalable engineering solutions.

EDUCATION

University of Maryland - College Park **GPA: 3.88/4**
Master of Engineering, Cloud Engineering Expected May 2026

- Cloud Computing, Cloud Security, Linux System Administration, Networks & Protocols, Data Storage & Databases

SKILLS

Programming Languages	Python, Bash/Shell Scripting, PowerShell, SQL, JavaScript, HTML, CSS
Cloud Services	**AWS:** IAM, EC2, Auto Scaling, Load Balancer, RDS, CloudWatch, CloudFront, S3, SNS, CloudTrail, WAF, Lambda, EKS, ECR **GCP:** Compute Engine, GKE (Kubernetes), Cloud Run, Cloud Functions, Cloud Storage, Cloud SQL, Pub/Sub, Cloud Monitoring, Cloud Deployment Manager **Azure:** Virtual Machines, Azure AD, Storage Accounts **Containers and orchestration:** Kubernetes, Docker, Helm **CI/CD:** GitHub Actions, GitLab CI/CD, Jenkins, CMake
Software & Tools	Terraform, Ansible, Prometheus, Grafana, ELK Stack, PostgreSQL, MySQL, pgAdmin4, Kafka, Git, VMware, VS Code, Linux, Databricks, REST APIs

WORK EXPERIENCE

Cloud Engineer *Newfold Digital, Mumbai, India* **Jun 23 – Jul 24**

- Supported **cloud infrastructure operations** for Linux-based production environments, resolving **7,000+ technical issues** across application, web, email, and backend service layers while maintaining a **4.84/5 CSAT** and **91.9% first-contact resolution**.
- Automated **infrastructure monitoring, health checks, and recurring operational workflows** using **Bash and Python**, reducing reactive incident volume by **15%** and improving overall service reliability and operational efficiency across customer systems.
- Supported **AWS environments, infrastructure configuration, & deployment validation**, gaining hands-on exposure to **Terraform, AWS CloudFormation, Docker, & Kubernetes-based environments** while contributing to **cloud platform operations**.
- Collaborated with engineering teams on **release support, incident triage, log analysis, and CI/CD-aligned operational tasks**, contributing to stronger platform stability in **cloud-native and microservices-oriented environments**.

Software Engineer Intern *Autumn Tech Labs, Mumbai, India* **Apr 22 – May 23**

- Developed and improved **backend features, RESTful APIs, and service-layer logic** using **Python**, increasing inventory processing efficiency by **25%** through workflow optimization, modular code improvements, and more efficient request handling.
- Debugged and resolved **application, service, and environment-level issues**, partnering with developers to improve deployment stability, reduce recurring defects, and strengthen backend reliability, maintainability, and overall system performance.
- Worked on **MySQL schema design, query tuning, and data integration workflows**, improving performance, strengthening data consistency, and ensuring reliable data access for core business operations and internal application workflows.
- Assisted with **testing, API integration, and cloud-supported deployment workflows**, gaining experience with **scalable software development, distributed application components, and modern deployment environments**.

PROJECTS

E-Commerce Microservices Platform on AWS EKS | *AWS, EKS, CloudFormation, Kubernetes, ALB, IRSA, GitHub Actions, Lambda*

- Provisioned a **private AWS EKS infrastructure** for a cloud-native e-commerce platform using **CloudFormation**, including VPC, subnets, route tables, NAT gateway, node groups, IAM roles, and ECR repositories.
- Deployed and configured **AWS Load Balancer Controller**, **IRSA**, and **ALB-based Ingress** for secure path-based routing across **6 containerized microservices**, enabling scalable and controlled access in a production-aligned Kubernetes environment.
- Integrated **Metrics Server**, **Cluster Autoscaler**, and **EBS CSI Driver** to support autoscaling, persistent volumes, and resilient workload scheduling, while enabling secure **CI/CD deployments** through **GitHub Actions** and a self-hosted bastion runner.
- Built a **serverless notification workflow** as part of the platform using **AWS Step Functions**, **SES**, **API Gateway**, **Lambda**, and **S3** to automate email reminders through secure endpoints with scalable, cost-efficient orchestration.

Customer Behavior Analytics Platform | *Python, PySpark, Databricks, Databricks SQL, Delta Lake, ETL*

- Built a **customer behavior analytics platform** using **Python, PySpark, and Databricks** to process user activity logs, transaction events, and application data for large-scale analytics, reporting, and customer usage pattern analysis.
- Developed **ETL pipelines, transformation workflows, & curated datasets** in **Databricks**, cleaning, aggregating, & enriching raw data to improve data quality & enable trend analysis, feature-level insights, operational reporting, and analytics use cases.
- Stored processed data in **Delta Lake** and queried it with **Databricks SQL**, enabling reliable analytics through ACID-compliant tables, schema enforcement, performant query execution, and scalable cloud-based data engineering workflows.

Predictive Infrastructure Auto-Scaler | *AWS EKS, CloudWatch, Amazon RDS, MySQL, Docker, Kubernetes, Prophet*

- Engineered a **predictive auto-scaling solution** deployed as a highly available **Dockerized workload** on **AWS EKS**, directly interfacing with the Kubernetes API to provision cluster capacity ahead of anticipated demand spikes, ensuring zero downtime.
- Developed a **time-series forecasting pipeline** utilizing **Prophet** machine learning model, trained on extensive historical CPU, memory, and request metrics to accurately identify underlying traffic trends and drive intelligent, automated scaling decisions.
- Integrated **Amazon RDS MySQL** for historical metric storage alongside **AWS CloudWatch** for real-time telemetry, enabling the continuous evaluation of predicted vs actual workload behavior to iteratively optimize forecasting accuracy & resource efficiency.